SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Dashboard Development Task

Course Code:	DSA05	**Course Title:**	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO5	**Assessment:**	Tool 1 – Dashboard Development Task
Weightage:	25%	**Total Marks:**	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
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Section / Batch:	AI&DS / 24 Batch	**Date of Submission:**	27-08-2026
Faculty In-charge:	K. Veena devi	**Project Mode:**	Individual Submission

Code of Conduct

I ___U.Ananya(192424382)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __U.Ananya_______

Student Academic Performance Dashboard

Objective

To develop a dashboard using Pandas, Matplotlib, and Seaborn to analyze student attendance, marks, assignments, laboratory performance, and final scores.

Dataset

The dataset contains:

- StudentID
- Attendance
- InternalMarks
- AssignmentMarks
- LabMarks
- FinalScore

Python Code

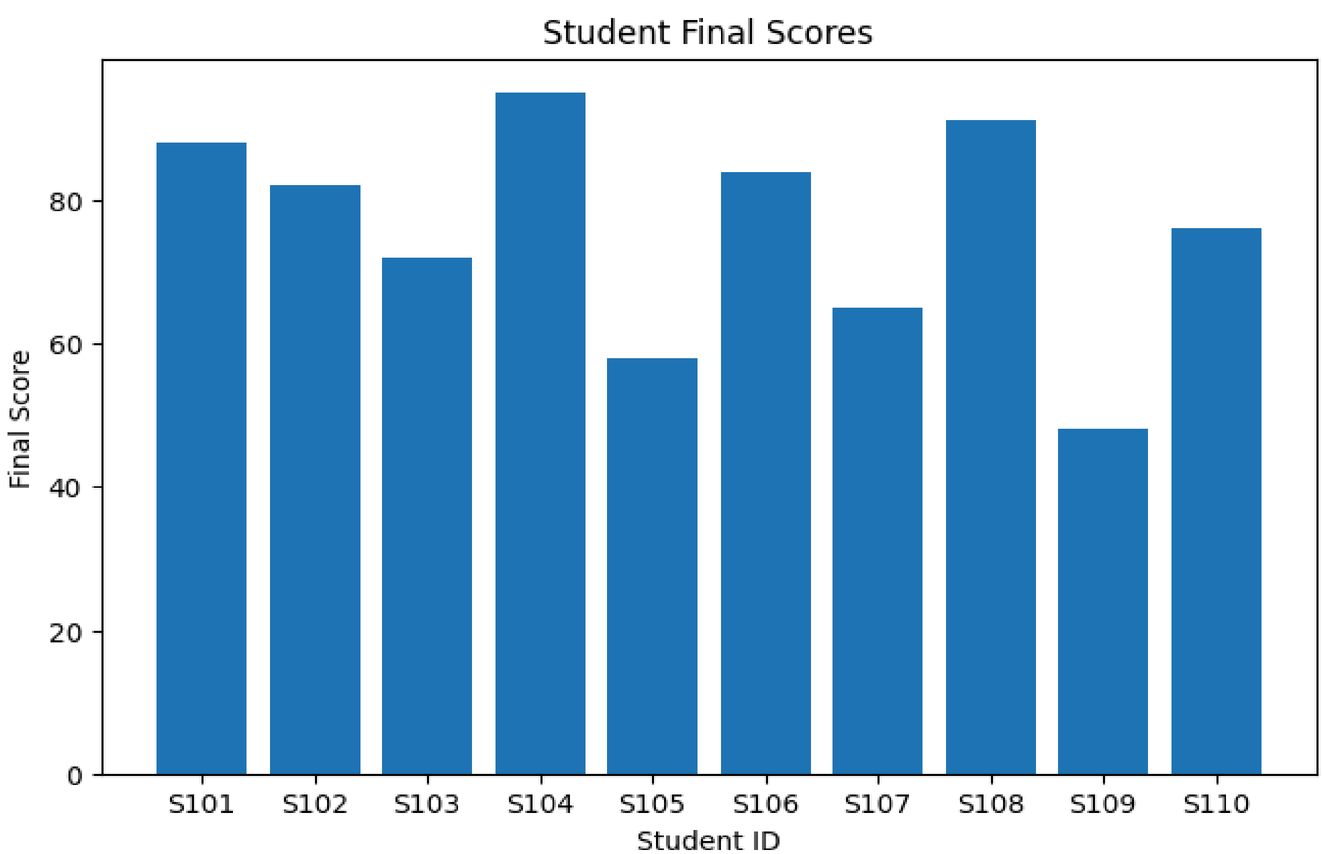
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("student_performance.csv")
print(df.head())
print(df.describe())
plt.figure(figsize=(8,5))
plt.bar(df["StudentID"], df["FinalScore"])
plt.title("Student Final Scores")
plt.xlabel("Student ID")
plt.ylabel("Final Score")
plt.show()
plt.figure(figsize=(8,5))
sns.histplot(df["FinalScore"], kde=True)
plt.title("Final Score Distribution")
plt.xlabel("Final Score")
```

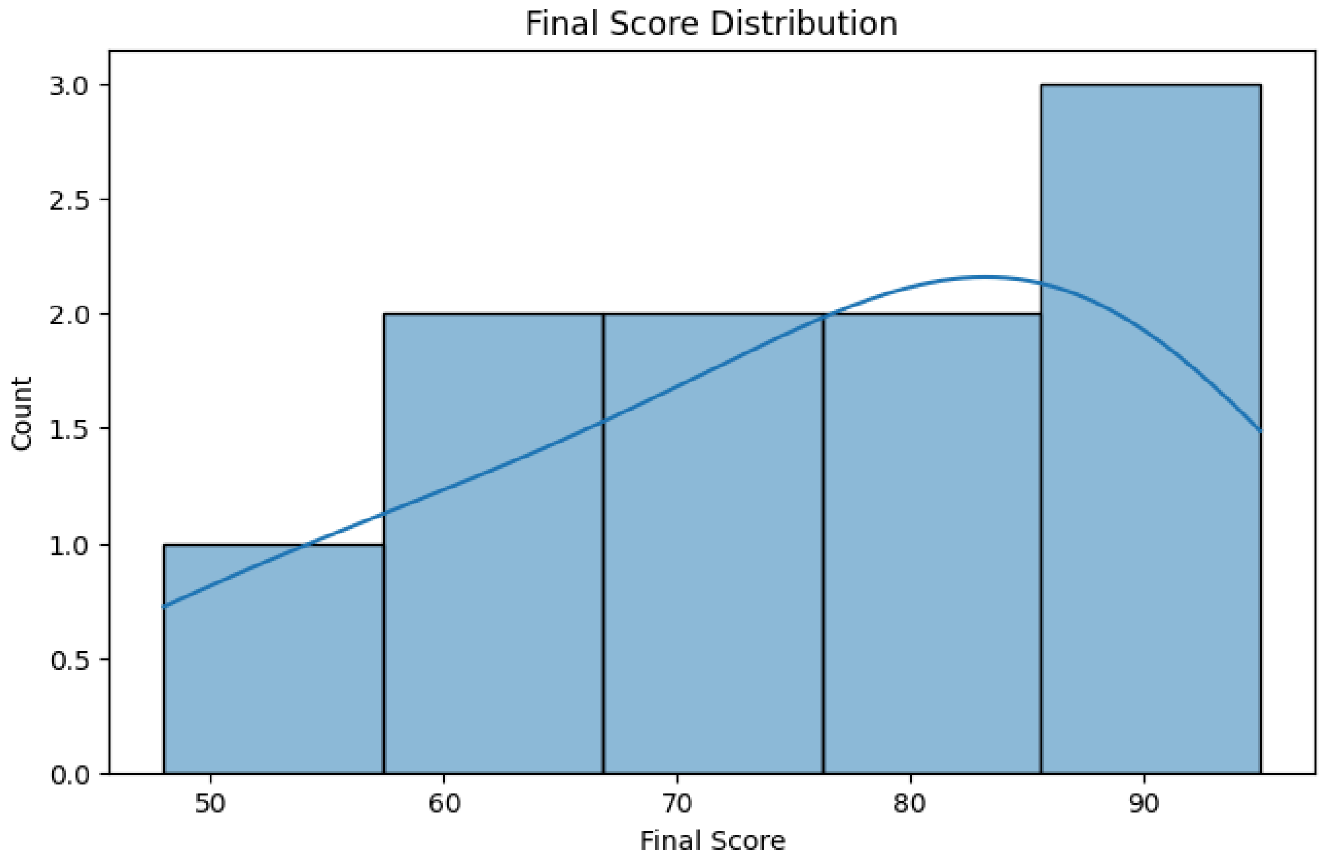
```
plt.show()
plt.figure(figsize=(8,5))
plt.scatter(df["Attendance"], df["FinalScore"])
plt.title("Attendance vs Final Score")
plt.xlabel("Attendance (%)")
plt.ylabel("Final Score")
plt.show()
plt.figure(figsize=(8,5))
sns.countplot(x="FinalScore", data=df)
plt.title("Final Score Frequency")
plt.show()
pd.plotting.scatter_matrix(
    df[[
        "Attendance",
        "InternalMarks",
        "AssignmentMarks",
        "LabMarks",
        "FinalScore"
    ]],
    figsize=(12,10)
)
plt.suptitle("Student Performance Scatter Matrix")
plt.show()
plt.figure(figsize=(8,6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm"
)
plt.title("Student Performance Correlation")
plt.show()
top_student = df.loc[df["FinalScore"].idxmax()]
```

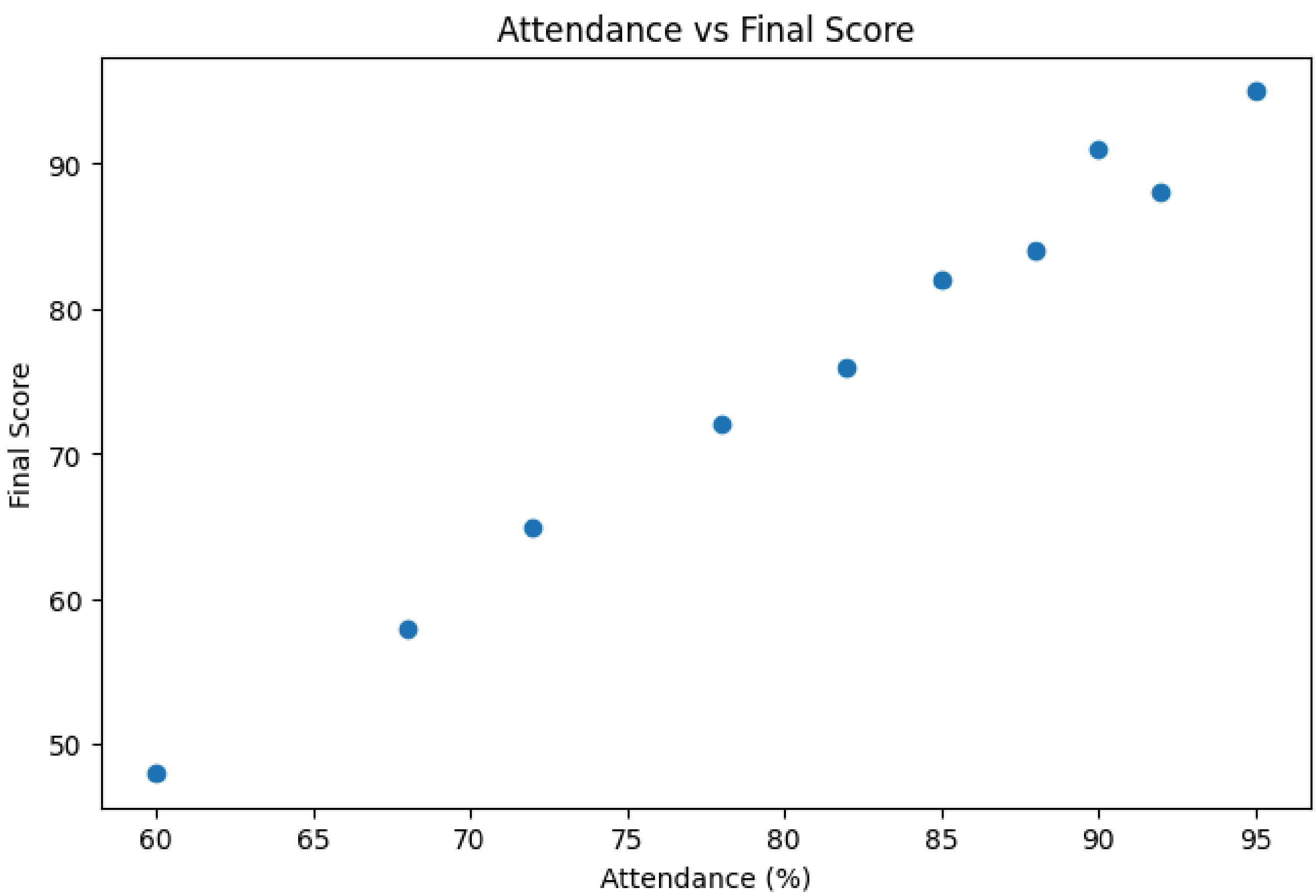
```
print("\nTop Performing Student:")
print(top_student)
print("\nStudents with Final Score below 50:")
print(df[df["FinalScore"] < 50])
```

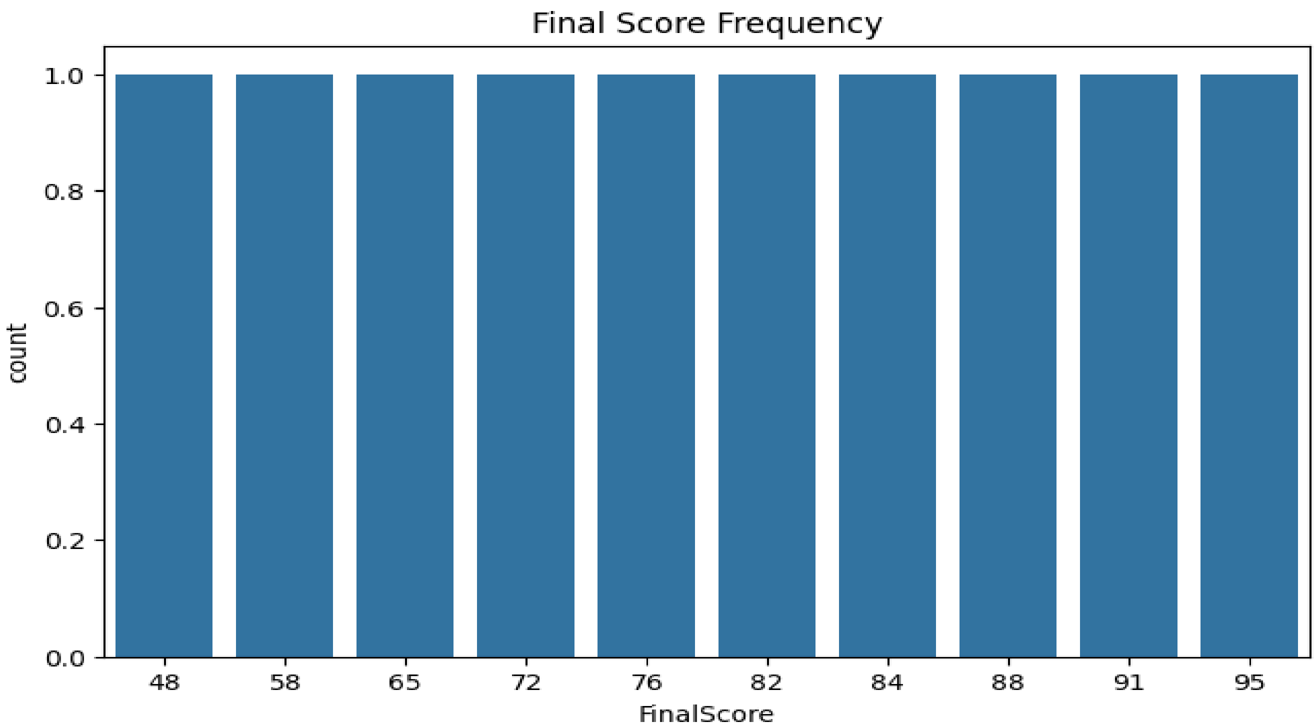
Dashboard Outputs:

```
...   StudentID  Attendance  InternalMarks  AssignmentMarks  LabMarks  FinalScore
0          S101          92             45               18        19          88
1          S102          85             40               17        18          82
2          S103          78             35               15        16          72
3          S104          95             48               19        20          95
4          S105          68             30               12        13          58
       Attendance  InternalMarks  AssignmentMarks   LabMarks  FinalScore
count   10.000000      10.000000        10.000000  10.000000    10.00000
mean    81.000000      38.100000        15.800000  16.600000    75.90000
std     11.372481       7.534366         3.047768   2.875181    15.18369
min     60.000000      25.000000        10.000000  11.000000    48.00000
25%     73.500000      32.750000        14.250000  15.250000    66.75000
50%     83.500000      39.000000        16.500000  17.500000    79.00000
75%     89.500000      44.250000        18.000000  18.750000    87.00000
max     95.000000      48.000000        19.000000  20.000000    95.00000
```

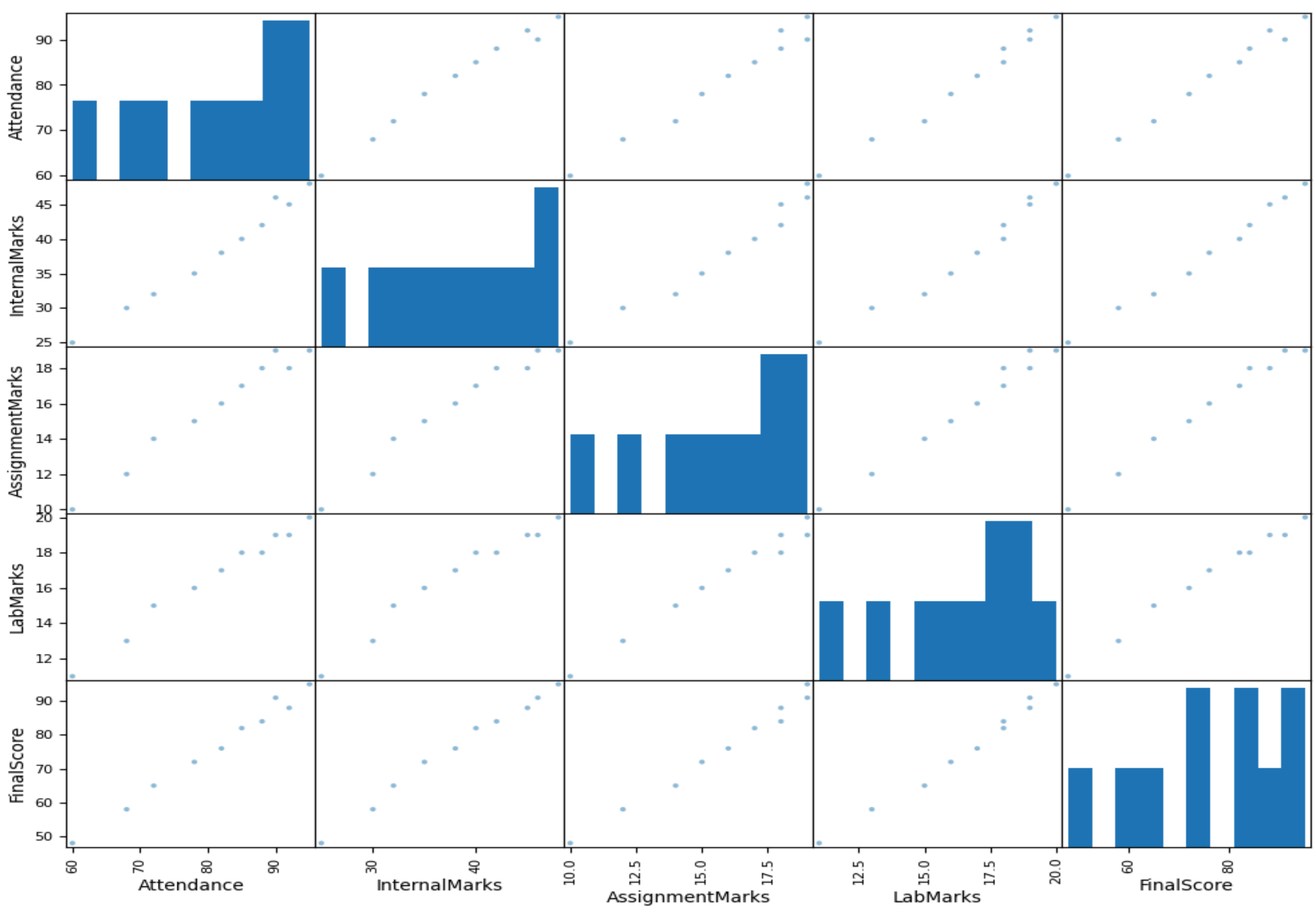


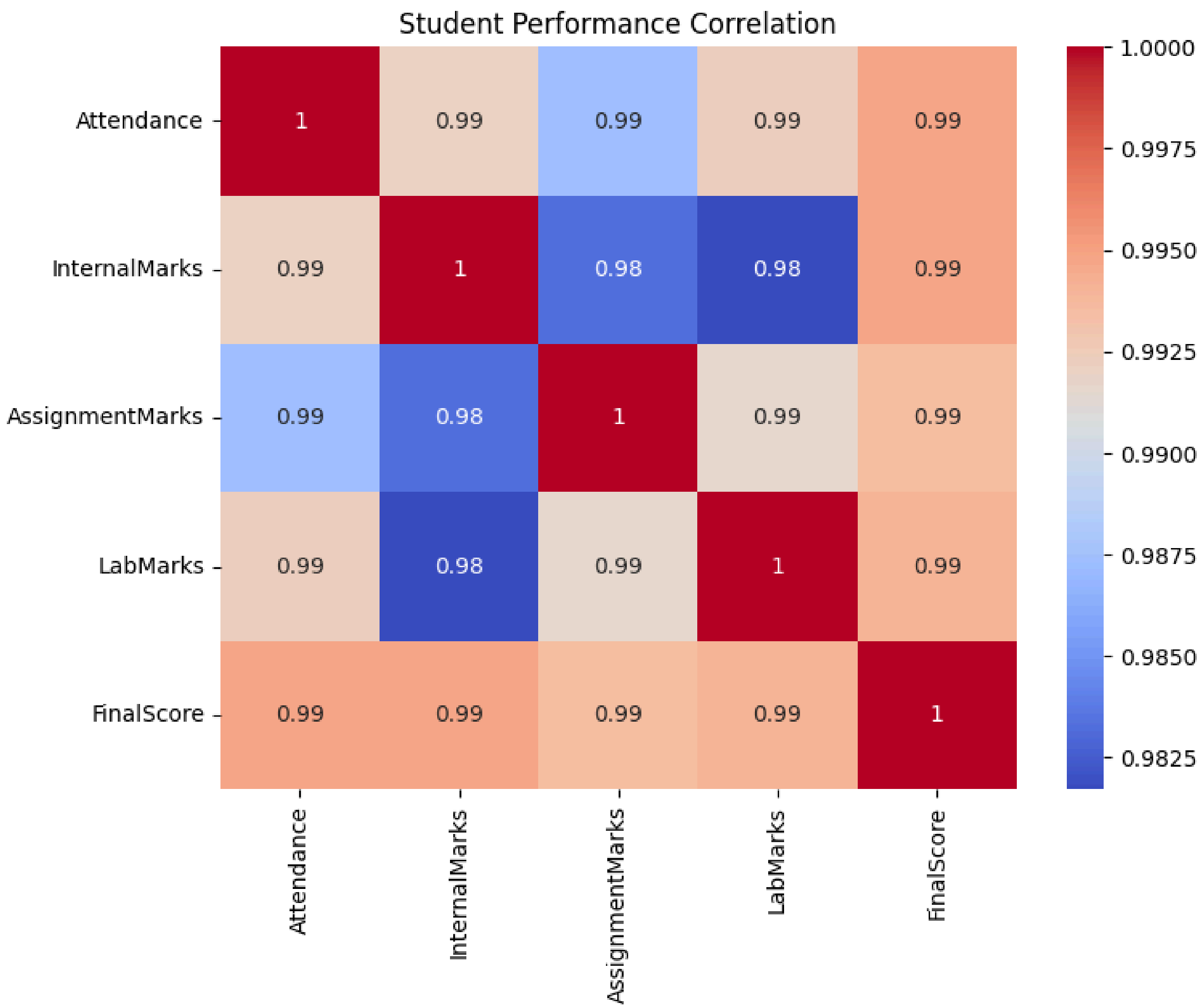

Final Score Distribution
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
50
60
70
80
90
Final Score


Attendance vs Final Score
Final Score
90
80
70
60
50
60
65
70
75
80
85
90
95
Attendance (%)



Student Performance Scatter Matrix






```
Top Performing Student:
StudentID          S104
Attendance           95
InternalMarks        48
AssignmentMarks      19
LabMarks             20
FinalScore           95
Name: 3, dtype: object

Students with Final Score below 50:
  StudentID  Attendance  InternalMarks  AssignmentMarks  LabMarks  FinalScore
8      S109          60             25               10        11          48
```

The program produces:

1. Bar Chart → Compares final scores of students.
2. Distribution Plot → Shows how final scores are distributed.

3. Scatter Plot → Shows the relationship between attendance and final score.
4. Frequency Chart → Shows the frequency of different scores.
5. Scatter Matrix → Shows relationships between attendance, internal marks, assignments, labs, and final scores.
6. Correlation Heatmap → Shows which academic factors are strongly related to final scores.

Conclusion

The Student Academic Performance Dashboard provides a simple way to understand student performance. It helps identify high-performing and low-performing students, study the relationship between attendance and marks, and understand how different academic factors contribute to final scores. The dashboard can help teachers identify students who may need additional academic support.